Teerth Patel

(832) 305-0722 | patelhteerth@gmail.com | linkedin.com/in/patelhteerth

EDUCATION

University of Texas at Austin Bachelor of Science, Electrical and Computer Engineering **May 2028**
GPA: 3.75/4.0

SKILLS/CERTIFICATIONS

Programming: Assembly, C, C++, JavaScript, Python, SQL, Linux, Java, HTML, CSS
Software: AutoCAD, MATLAB, Vivado, KiCAD, LTspice, Altium Designer, Microsoft Office, Google Suite
Hardware: Multi-layer PCB design, schematic capture & layout, digital/analog prototyping, SMT & reflow soldering
Certifications: E2A Soldering, Autodesk Inventor
Coursework: Circuit Theory, Digital Logic Design, VLSI Design, Embedded Systems, Signal and Systems

EXPERIENCES

Corveus Medical - Electrical Engineering Intern **May 2026 - Present**

- Designed a mixed-signal waveform-generation architecture integrating a 16-bit DAC, digital SPI control, isolated ±15 V power, precision signal conditioning, and a Howland current-source output stage, connecting digital control circuitry with precision analog hardware.
- Developed Python/FTDI SPI control and validation software for the DAC, automating 16-bit command generation, device initialization, waveform updates, and output verification during hardware bring-up.

Rice University - Undergraduate Researcher, ECE **May 2026 - Present**

- Integrated TI AWR1843 and Analog Devices TinyRad radar systems with ZED stereo vision on an NVIDIA Jetson Xavier NX, coordinating multiple digital sensor interfaces within an embedded computing platform.
- Developed Python-based acquisition and synchronization routines and validated 300+ synchronized frames through structural and timestamp checks, ensuring reliable communication between sensing and processing subsystems.

PSMS Controls Services - Hardware Engineering Intern **Jan 2026 - Present**

- UART, SPI, and RS-485 digital communication interfaces using oscilloscopes and logic analyzers, resolving 20+ timing, grounding, and signal-integrity faults during system commissioning.
- Debugged and validated more than 50 analog and digital I/O channels, analyzing interface behavior, power distribution, filtering, and transient-protection circuitry across embedded control hardware.

PROJECTS

FPGA RTL Stopwatch – Xilinx Basys 3 **Feb 2026 - Apr 2026**

- Designed a modular Verilog RTL architecture implementing finite-state control, synchronous counters, clock division, input debouncing, and seven-segment display logic on a Xilinx Artix-7 FPGA.
- Implemented the design in Vivado, integrating sequential and combinational logic modules into a functional FPGA system with hardware-controlled start, stop, reset, and timing functionality.

LEADERSHIP EXPERIENCE AND ACTIVITES

Longhorn Neurotech - Hardware Systems Team **Sep 2025 - Present**

- Integrated distributed digital interfaces across LiDAR, RGB, ultrasonic, and embedded-control subsystems, coordinating hardware and software timing to maintain ≤5 ms synchronization across system modules.
- Developed motor-control interface hardware supporting a 500 Hz closed-loop control architecture, translating system-level requirements into reliable embedded hardware interfaces.

Longhorn Baja - Research & Development - Electronics Lead **Aug 2025 - Present**

- Led debugging and validation across 15+ vehicle electrical subsystems, diagnosing communication faults, grounding problems, connector failures, and voltage irregularities during system testing.
- Standardized electrical validation procedures and troubleshooting workflows, contributing to a 30% increase in vehicle uptime during competition endurance events.